

Day 19 Quiz

Geometry Summer Credit | Arc Length, Sector Area, and Radian Readiness

Name:	Date:	Score:
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Directions

Choose the best answer. Show the fraction of a circle for short-response items.

1.	A circle has radius 8 and central angle 90 degrees. The arc length is: A 2 pi B 4 pi C 8 pi D 16 pi
2.	A circle has radius 6 and central angle 120 degrees. The sector area is: A 6 pi B 12 pi C 24 pi D 36 pi
3.	A 45-degree central angle represents what fraction of a circle? A 1/4 B 1/6 C 1/8 D 1/45
4.	180 degrees equals: A pi/2 radians B pi radians C 2 pi radians D 180 pi radians
5.	3 pi/4 radians equals: A 45 degrees B 90 degrees C 135 degrees D 270 degrees
6.	Find the arc length for radius 15 and central angle 144 degrees.
7.	Find the sector area for radius 10 and central angle 72 degrees.

Day 19 Quiz

Continued

8.	A circle has radius 18 and arc length 6π . Find the central angle.
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9.	Convert 225 degrees to radians.
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10.	Spiral review: A central angle and an inscribed angle intercept the same 160-degree arc. Find both angles.
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11.	Error analysis: A student uses $(\theta/360)(2\pi r)$ to find sector area. Explain the error and write the correct formula.
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